Ayush Jain

Research Scientist
Applied Reinforcement Learning
Meta

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Google Scholar
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Research & Impact

- **LLM post-training & agents:** Production and research experience in SFT, GRPO, multi-turn evaluation, credit assignment, self-improvement, and computer-use agents.
- **Reinforcement learning:** Research on decision making in Complex Action Spaces **(thesis)**, multi-task RL, and robot learning, with publications at ICML, ICLR, NeurIPS, EACL, and RLC; recipient of the **RLC 2025 Outstanding Paper Award**.
- **Environment & benchmark design:** Developed **CREATE** for action-space generalization, **Restricted Mujoco** for continuous-control failure modes, and **BayesBench** for belief estimation in multi-turn LLM interactions.

Professional Experience

Meta, Sunnyvale, CA — Feb 2025 – Present
Research Scientist, Applied Reinforcement Learning Team, Monetization

- **Post-trained production LLM conversation agents** using SFT and GRPO; developed multi-turn LLM-judge evaluation for Messenger Business Messaging.
- **Lead modeling for multi-step RL in ad recommendation**, applying CQL, IQL, and TD3-BC at industrial scale across retrieval and ranking.
- Developed **large-scale training and evaluation infrastructure** for offline RL models across the production recommendation stack at both retrieval and ranking stages.

Meta Reality Labs Research, Redmond, WA — May 2024 – Dec 2024
Research Intern with Nitin Kamra

- Trained VLM-based **computer-use agents** with advantage-weighted regression (AWR) for natural-language instruction following in Android agentic tasks.

Microsoft Research, Montreal, Canada — May 2023 – Jan 2024
Research Intern with Eric Yuan and Marc-Alexandre Côté

- Discrete prompt optimization for LLM-based instruction-following agents in interactive Minecraft environments.

Naver AI Research, Seoul, South Korea (remote) — 2021, 2022
Research Intern with Kyung-Min Kim and Joseph J. Lim

- RL for recommender systems with large, varying, and slate action spaces.

Samsung Research, Seoul, South Korea — Sep 2016 – Jun 2018
Engineer, Data Analytics Lab with James Geraci and Yunsu Lee

- Time series modeling for (i) market share prediction and (ii) anomaly detection.

Education

University of Southern California — Dec 2024
PhD and MS in Computer Science (Advisors: Joseph J. Lim and Erdem Bıyık) — GPA: 4.0/4.0

Indian Institute of Technology Delhi — Jun 2016

B.Tech in Electrical Engineering

Publications

Peer-Reviewed Conference Papers

[C9] **NeurIPS 2026**. A. Samanta, A. Magesh, **A. Jain**, Y. Yu, D. Jiang, K. Asadi, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. "Credit Assignment with Resets in Language Model Reasoning". *Neural Information Processing Systems.*

[C8] **ICLR 2026**. X. Li, **A. Jain**, Z. Yang, Y. Korkmaz, E. Bıyık. "When a Robot is More Capable than a Human: Learning from Constrained Demonstrators". *International Conference on Learning Representations.*

[C7] **ICML 2026**. A. Samanta, A. Magesh, **A. Jain**, K. Asadi, Y. Yu, D. Jiang, B. Vidolov, K. Hassani, P. Sajda, J. Bhandari, Y. Efroni. "Structure Enables Effective Self-Localization of Errors in LLMs". *International Conference on Machine Learning.*

[C6] **EACL 2026**. R. Wu, A. Samanta, **A. Jain**, S. Fujimoto, J. Kwon, B. Kretzu, Y. Yu, K. Hassani, B. Vidolov, Y. Efroni. "Imbalanced Gradients in RL Post-Training of Multi-Task LLMs". *19th Conference of the European Chapter of the Association for Computational Linguistics.*

[C5] **NeurIPS 2025**. Y. Korkmaz*, U. Bhuwania*, **A. Jain**†, E. Bıyık†. "Actor-Free Continuous Control via Structurally Maximizable Q-Functions". *Neural Information Processing Systems.*

[C4] **RLC 2025**. **A. Jain**, N. Kosaka, X. Li, K. Kim, E. Bıyık, J. Lim. "Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions". *Reinforcement Learning Conference.* **Outstanding Paper Award on Empirical Reinforcement Learning Research**.

[C3] **ICLR 2025**. G. Zhang*, **A. Jain***, I. Hwang, S. Sun, J. Lim. "QMP: Q-switch Mixture of Policies for Multi-Task Behavior Sharing". *International Conference on Learning Representations.*

[C2] **ICLR 2022**. **A. Jain***, N. Kosaka*, K. Kim, J. Lim. "Know Your Action Set: Learning Action Relations for Reinforcement Learning". *International Conference on Learning Representations.*

[C1] **ICML 2020**. **A. Jain***, A. Szot*, J. Lim. "Generalization to New Actions in Reinforcement Learning". *International Conference on Machine Learning.*

Selected Workshop Papers

[W3] **RSS 2026 Workshop**. R. Lindeborg, R. J. Peng, **A. Jain**, J. Zhang, A. Anwar, J. Thomason. "Anticipating Unintended Robot Behaviors with Side Effect Critics". *Rethinking What It Means to be "Safe" for Generalist Robots.* **Spotlight presentation**.

[W2] **COLING 2018 Workshop**. **A. Jain**, V. Singh, S. Ranjan, R. Rajkumar, S. Agarwal. "Uniform Information Density Effects on Syntactic Choice in Hindi". *Workshop on Linguistic Complexity and NLP.*

[W1] **DGfS 2017**. **A. Jain**, V. Singh, S. Agarwal, R. Rajkumar. "Uniform Information Density models for language production". *39th Annual Conference of the German Linguistic Society.*

Preprints

[P2] arXiv. A. Samanta, A. Magesh, T. Lancewicki, **A. Jain**, Y. Yu, P. Sajda, K. Hassani, A. Modi, D. Jiang, Y. Efroni. "BayesBench: Evaluating LLM Belief Trajectories Under Multi-Turn Evidence Accumulation".

[P1] arXiv. A. Samanta, A. Magesh, R. Wu, **A. Jain**, Y. Yu, D. Jiang, B. Vidolov, P. Sajda, Y. Efroni, K. Hassani. "Self-Improvement of Language Models by Post-Training on Multi-Agent Debate".

Patent

[T1] J. Lee, M. Kim, **A. Jain**, T. Hwang, J. Kim, H. Cho. "Method and Apparatus for Managing Operation Data of Appliance for Failure Prediction". U.S. Patent No. 11,182,235. Nov 23, 2021.

Research Mentoring

Norio Kosaka (Naver AI → Ph.D. student at NII Japan)	ICLR 2022, RLC 2025
Xinhu Li (USC → Ph.D. student at NUS)	RLC 2025, ICLR 2026
Urvi Bhuwania (USC)	NeurIPS 2025
Zhaoyi Zhou (CMU, Meta research intern)	Current

Technical Expertise

Large scale reinforcement learning

- LLM post-training (SFT, GRPO), multi-turn LLM judge evaluation for Business Messaging.
- Offline RL for Ad RecSys (CQL, IQL, TD3-BC).
- LLM judge and AWR for computer-use android agent.

Systems:

- Distributed training for recommendation systems: retrieval, ranking, and auction.

Teaching and Service

Teaching Assistant, University of Southern California

Deep Learning and its Applications, CSCI-566.

- Fall 2019; Spring 2019; Fall 2020; Spring 2023; Spring 2024; Fall 2024.

Reviewer Service

ICLR (2023–2027); NeurIPS (2023–2026); NeurIPS Workshop Proposals (2026); ICML (2025–2026); CoRL (2021–2024); RLC (2025–2026); AAAI (2026).

Selected Honors

- Outstanding Paper Award on Empirical Reinforcement Learning Research (RLC 2025).
- Silver Reviewer Award, International Conference on Machine Learning (ICML 2026).